Digital System Design

N16 ADFP Guidelines

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Outline

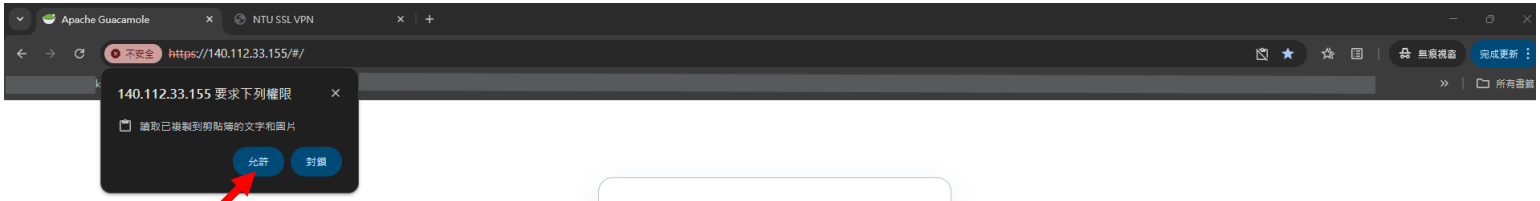
- Workstation Connection
- Synthesis with N16 ADFP

VPN

- Download Ivanti Secure Access Client and set up the connection to NTU VPN Server
- Follow the steps of NTU SSL VPN Services
- Make sure you connect to NTU VPN while using ADFP workstation, **even if you are under the NTU network**

Remote Desktop Connection

- Open browser and connect to 140.112.33.155
- Don’t change the password

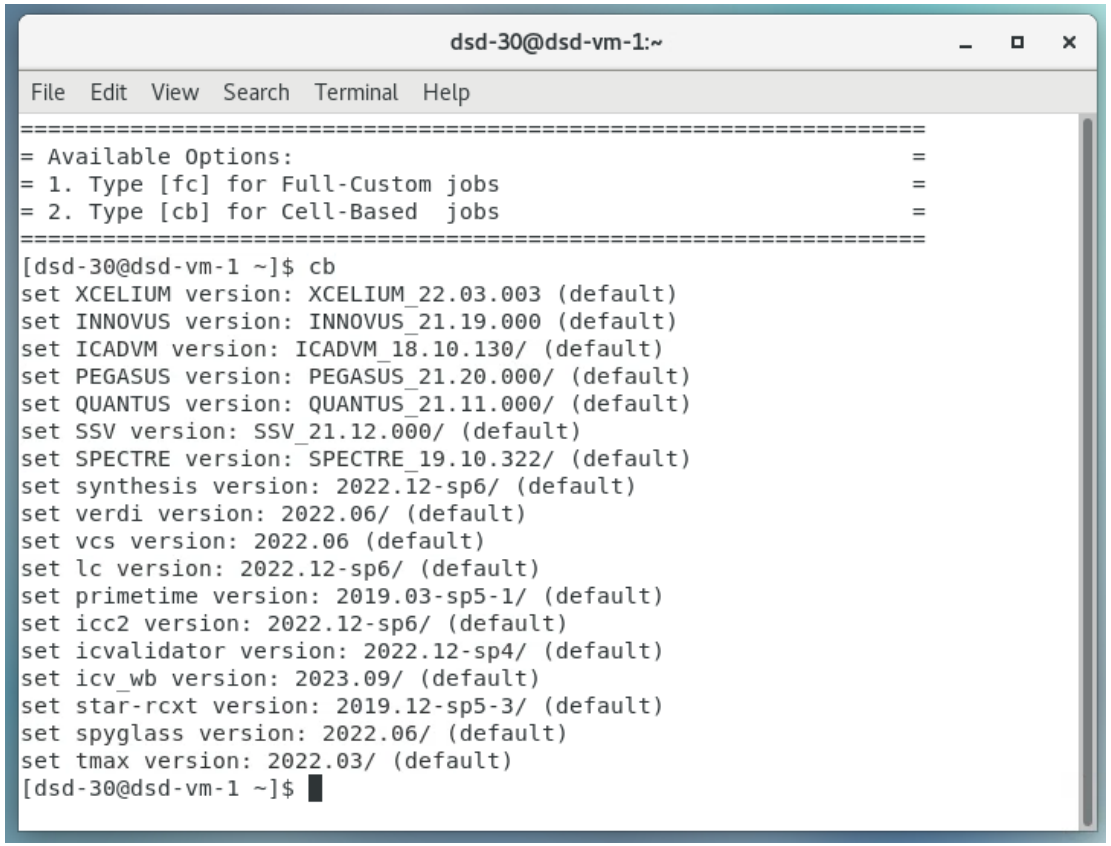


1. **Agree to allow copy content**

2. **Fill in your account(dsd-xx) and your password**

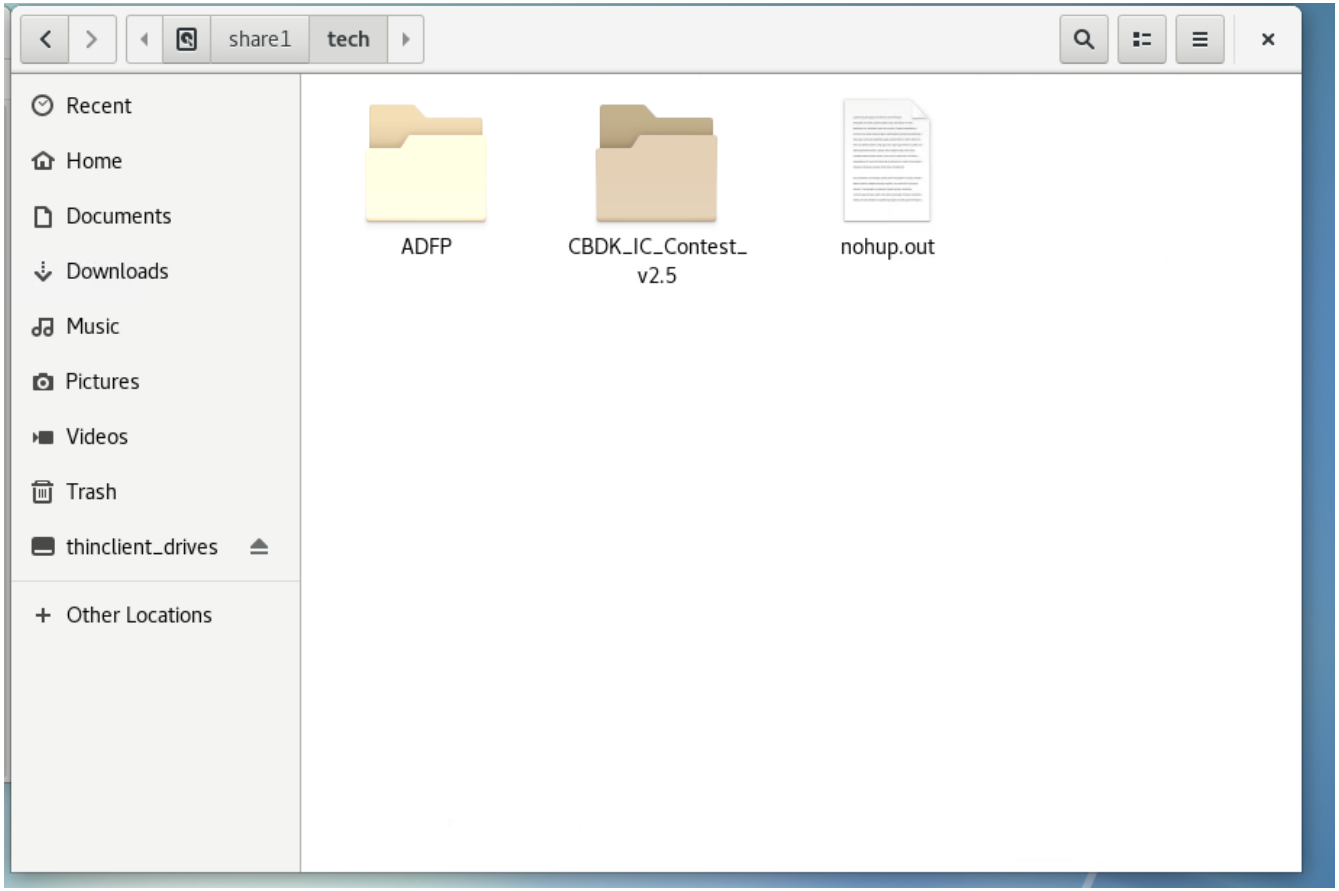
Remote Desktop Connection

- Open terminal and type “cb” to source the license



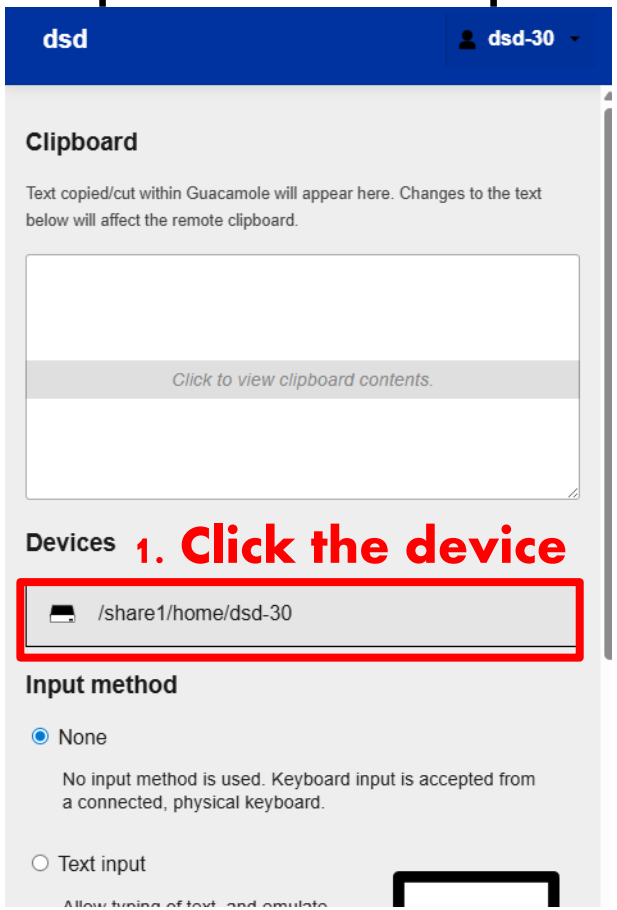
ADFP Library

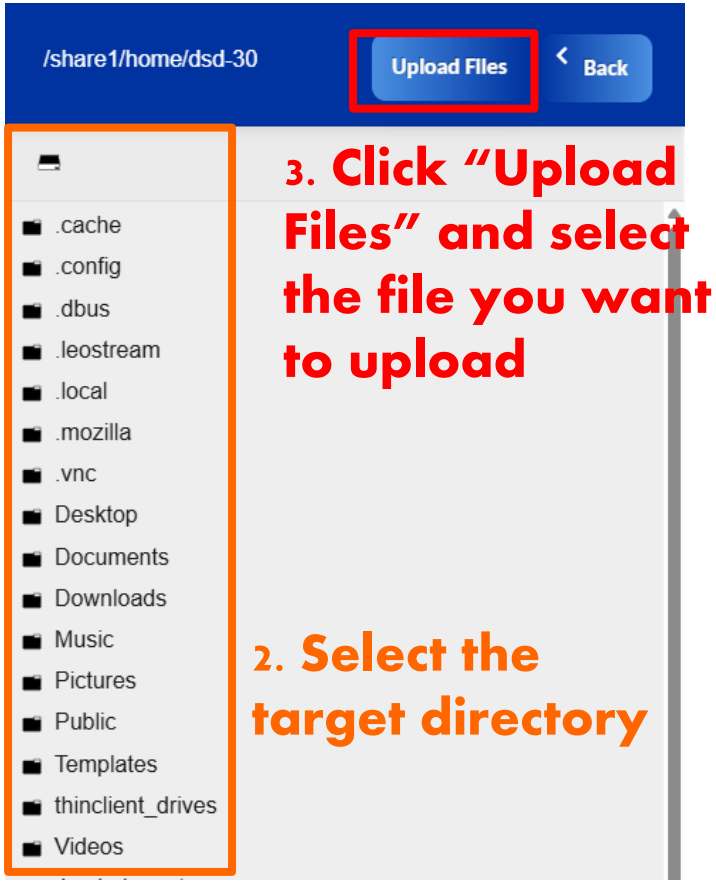
- ADFP library: /share1/tech/ADFP



Upload Files to Workstation

- Open control panel by **ctrl+shift+alt**



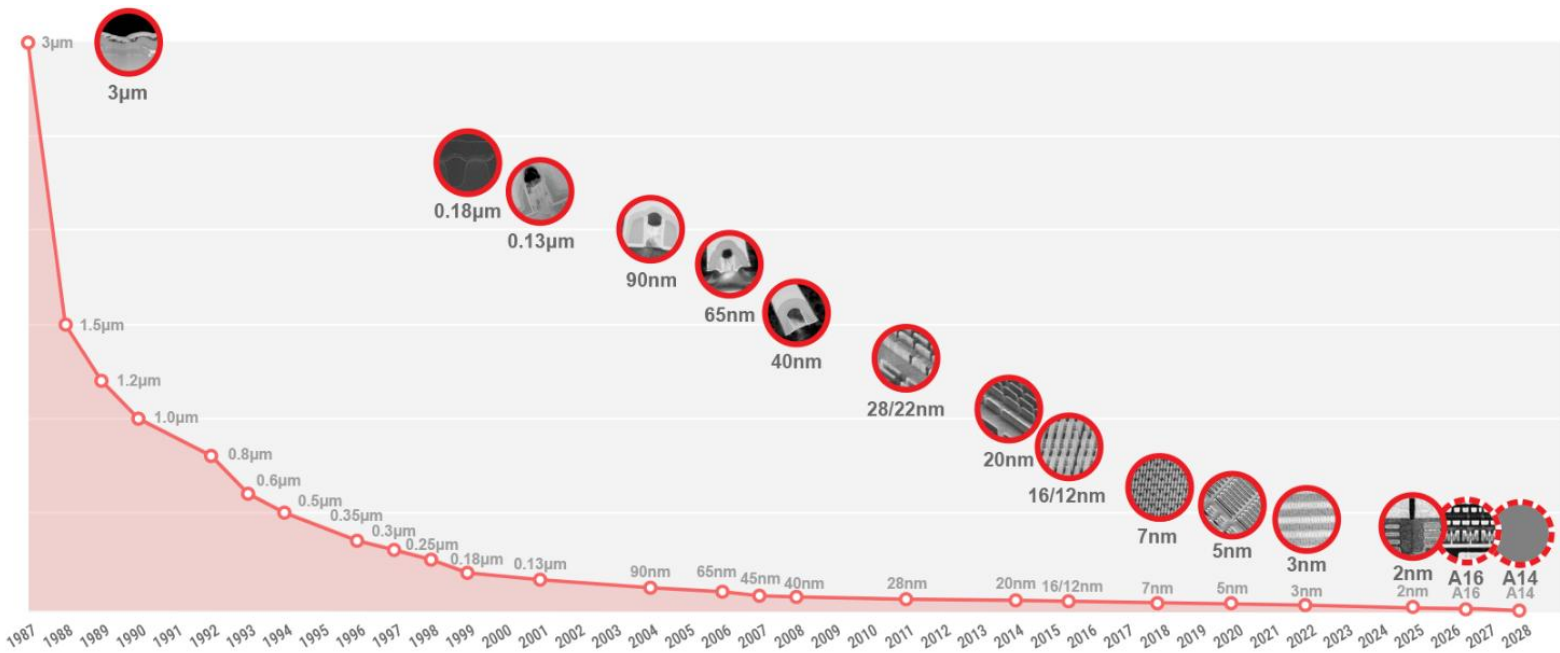


Workstation FAQ

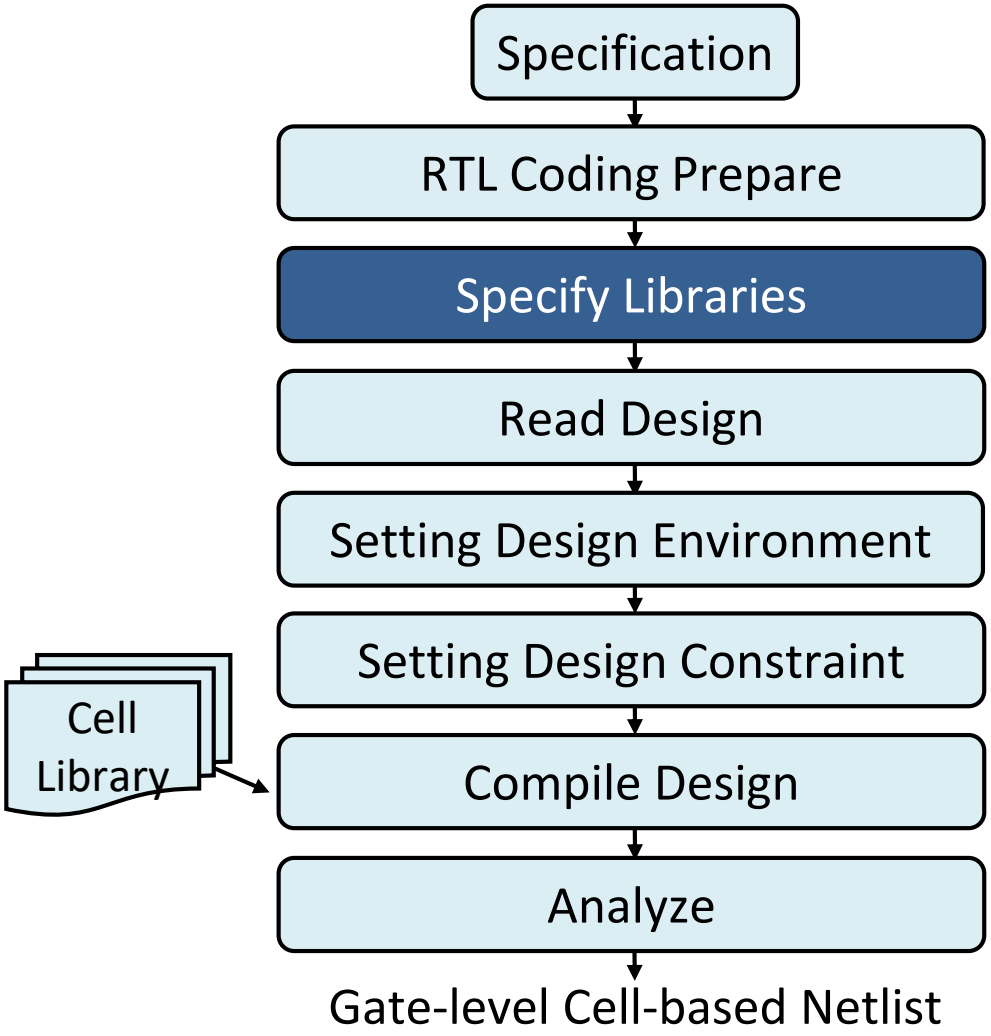
- Join the Discord server to get real-time workstation updates and ask related questions
- https://discord.gg/ZPZqA2t6AF

TSMC ADFP Introduction

- Academic Design Forster Package (ADFP) is a virtual 16nm FinFET CMOS
- Technology developed by TSMC for teaching purposes only
  - Full-custom & standard-cell
  - much more design rules



Synthesis with ADFP



Cell Library in N16 ADFP

- **P**rocess: fast/slow/typical corner
- **V**oltage: 0.72V/0.8V/0.88V
- **T**emperature: -40 ~ 125˚C

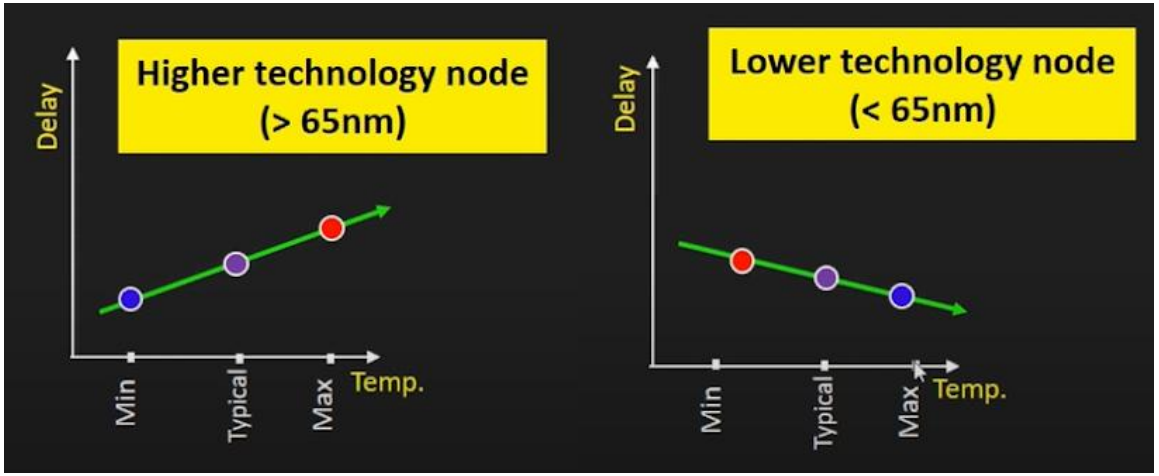
```
set search_path     "/share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/ \
                     $search_path .\
                     "

set target_library  "N16ADFP_StdCellff0p88v125c_ccs.db \
                     N16ADFP_StdCellff0p88vm40c_ccs.db \
                     N16ADFP_StdCellss0p72v125c_ccs.db \
                     N16ADFP_StdCellss0p72vm40c_ccs.db \
                     N16ADFP_StdCelltt0p8v25c_ccs.db \
                     "
```

In 0.13μm: slow.db / fast.db

Temperature Inversion

- When T increase, both μ_n and V_t increase
- In advanced technology, the second factor dominated because V_{gs} and V_t is closer
- Multi-Corner Analysis is needed

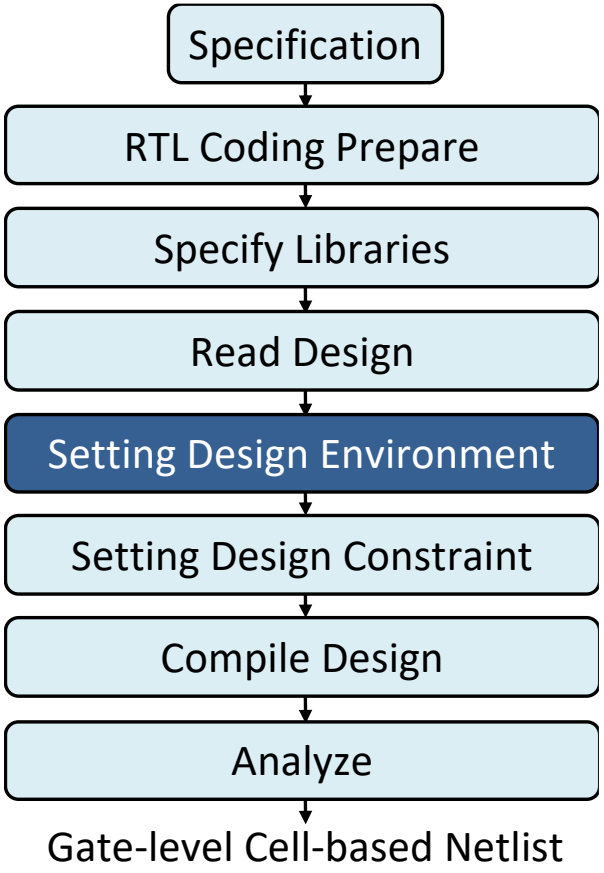


$$I_d = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left(V_{gs} - V_t\right)^2$$

Design Environment

- Net delay provided by wire load model is not accurate
- No wire load model in ADFP

1. set_input_delay
2. set_output_delay
3. set_drive
4. set_load
5. set_operating_conditions
6. ~~set_wire_load_model~~



Hold Time Violation

- Hold time violation is quite common as the circuit is very fast

Check if there are negative slack

```
report_timing -delay max -max_paths 5
```

```
report_timing -delay min -max_paths 5
```

```
Startpoint: x_r_reg_1__2_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: x_r_reg_1__2_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
GSIM               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                        Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                        0.00       0.00
clock network delay (ideal)                  0.50       0.50
x_r_reg_1__2_/CP (DFCNQD2BWP16P90LVT)        0.00       0.50 r
x_r_reg_1__2_/Q (DFCNQD2BWP16P90LVT)         0.03       0.53 f
U3358/ZN (ND2D1BWP20P90)                     0.01       0.54 r
U3056/ZN (OAI21D1BWP20P90)                   0.01       0.54 f
x_r_reg_1__2_/D (DFCNQD2BWP16P90LVT)         0.00       0.54 f
data arrival time                                       0.54

clock clk (rise edge)                        0.00       0.00
clock network delay (ideal)                  0.50       0.50
clock uncertainty                            0.10       0.60
x_r_reg_1__2_/CP (DFCNQD2BWP16P90LVT)        0.00       0.60 r
library hold time                            0.01       0.61
data required time                                      0.61
-----------------------------------------------------------
data required time                                      0.61
data arrival time                                      -0.54
-----------------------------------------------------------
slack (VIOLATED)                                       -0.06

slack (VIOLATED: increase significant digits)                0.00
```

Compile Design

- May still have hold time violations after compile_ultra

```
set_fix_hold [all_clocks]
```

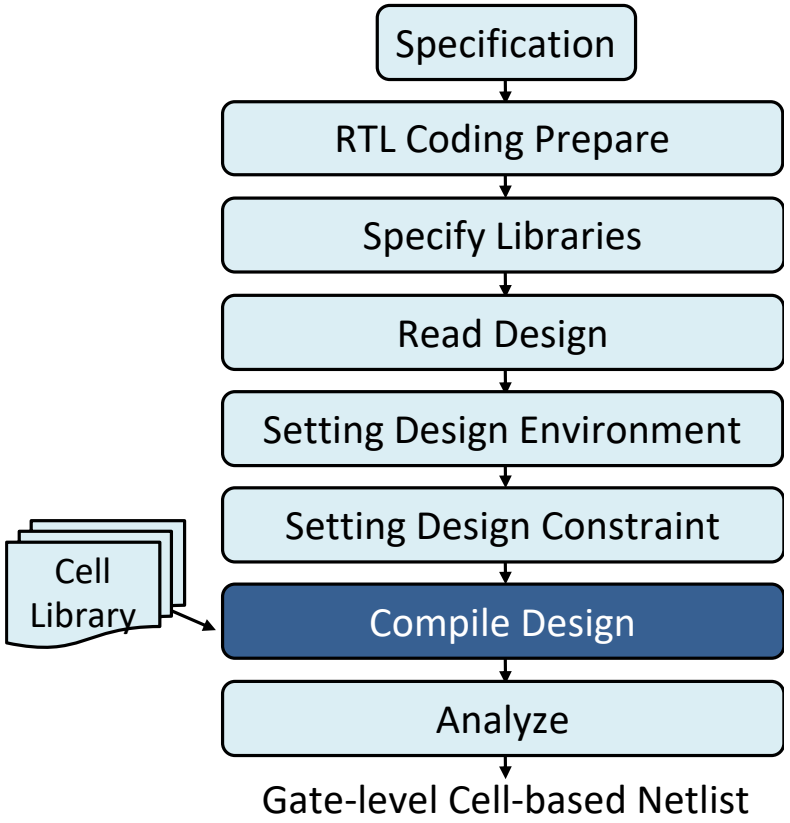
```
compile_ultra
```

+

```
compile_ultra -inc
```

+

```
compile –inc –only_hold_time
```



Compile Design

- No preset pin in ADFP register

```
always @(posedge clk or posedge rst) begin
    if (rst) begin
        in_a_buffer <= 1;
end
    else begin
        in_a_buffer <= in_a_buffer_nxt;
    end
end
```

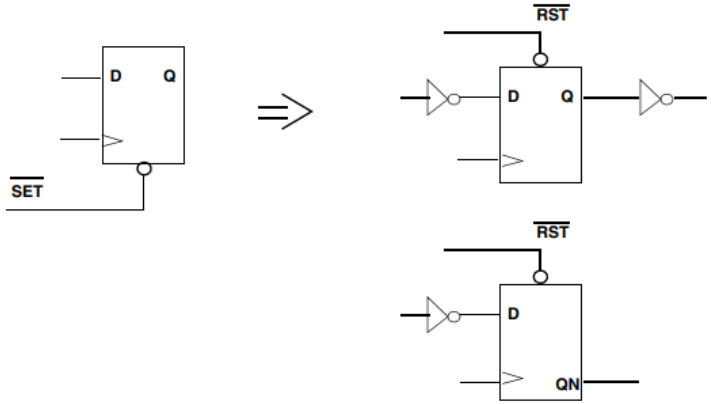
```
CKND1BWP16P90LVT U28 ( .I(rst_n), .ZN(n18) );
\**FFGEN**  in_a_buffer_reg_0_ ( .next_state(n9), .clocked_on(clk),
      .force_00(net434), .force_01(net434), .force_10(rst_n), .force_11(
      net434), .Q(in_a_buffer[0]), .QN(n20) );
\**FFGEN**  in_b_buffer_reg_0_ ( .next_state(in_b[0]), .clocked_on(clk),
      .force_00(net434), .force_01(net434), .force_10(rst_n), .force_11(
      net434), .Q(in_b_buffer[0]) );
TIELBWP20P90LVT U33 ( .ZN(net434) );
```

Compile Design

- Set the option

```
set compile_seqmap_enable_output_inversion true
```

```
CKND1BWP16P90LVT U28 ( .I(rst_n), .ZN(n18) );
\**FFGEN**  in_a_buffer_reg_0_ ( .next_state(n9), .clocked_on(clk),
      .force_00(net434), .force_01(net434), .force_10(rst_n), .force_11(
      net434), .Q(in_a_buffer[0]), .QN(n20) );
\**FFGEN**  in_b_buffer_reg_0_ ( .next_state(in_b[0]), .clocked_on(clk),
      .force_00(net434), .force_01(net434), .force_10(rst_n), .force_11(
      net434), .Q(in_b_buffer[0]) );
TIELBWP20P90LVT U33 ( .ZN(net434) );
```




```
DFCNQD2BWP16P90LVT out_reg_0_ ( .D(N1), .CP(clk), .CDN(n12), .Q(out[0]) );
DFCNQND1BWP16P90LVT in_b_buffer_reg_0_ ( .D(n14), .CP(clk), .CDN(n12), .QN(
      in_b_buffer[0]) );
DFCNQND1BWP16P90LVT in_a_buffer_reg_0_ ( .D(n30), .CP(clk), .CDN(n12), .QN(
      in_a_buffer[0]) );
CKND1BWP16P90LVT U15 ( .I(enable), .ZN(n13) );
```

Thanks! Feel free to ask me any questions!